Report R03 — FASN and the Lipogenic Phenotype in HCC

Independent test of the withdrawn manuscript's three claims.

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Summary

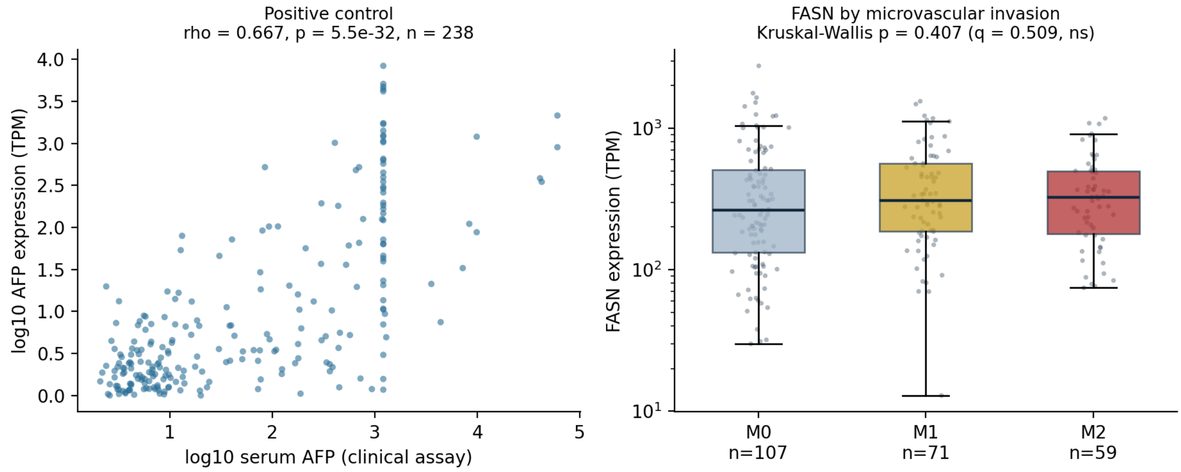
The withdrawn HCC manuscript claimed FASN associates with tumour stage, vascular invasion and survival. We tested all three in the Chinese Liver Cancer Atlas (238 RNA-sequenced tumours), a cohort independent of the TCGA-LIHC data in which those claims previously failed. **None of the three associations is present** after multiplicity correction.

What *does* replicate is the lipogenic phenotype itself: FASN co-expresses tightly with ACACA ($\rho = +0.546$), SCD (+0.510) and SREBF1 (+0.495), all $p < 4 \times 10^{-16}$.

A pre-registered positive control confirms the analysis is sound rather than merely negative: AFP gene expression correlates with the independently measured serum AFP concentration at $\rho = 0.667$, $p = 5.5 \times 10^{-32}$ — two different assays of the same biology, agreeing strongly. The clinical join and the expression matrix are therefore verified.

1 Results

Test	**Statistic**	p	q **(BH)**
Positive control			
AFP expression vs serum AFP ($n = 238$)	$\rho = 0.667$	5.5×10^{-32}	—
Lipogenic co-expression (replication)			
FASN ~ ACACA	$\rho = +0.546$	7.1×10^{-20}	—
FASN ~ SCD	$\rho = +0.510$	3.8×10^{-17}	—
FASN ~ SREBF1	$\rho = +0.495$	3.9×10^{-16}	—
The manuscript's claims			
Vascular invasion (M0/M1/M2, $n = 237$)	$H = 1.80$	0.407	0.509
absent vs present	—	0.201	—
BCLC stage	$H = 1.62$	0.656	0.656
Edmondson grade	$H = 2.39$	0.122	0.299
Survival status (living vs deceased, $n = 183$)	—	0.179	0.299
Recurrence status ($n = 183$)	—	0.072	0.299



2 Interpretation, with the nuance the numbers require

The lipogenic phenotype is real. FASN sits in a coherently co-regulated lipogenic programme, replicated here and previously in TCGA-LIHC ($\rho = +0.62$, $+0.62$, $+0.42$). That part of the manuscript survives.

The clinical associations do not. Two independent cohorts now agree: FASN expression does not track stage, vascular invasion or outcome.

But this is absence of evidence for a *small* effect, not proof of no effect. The audit quantified what we could have detected. For vascular invasion absent versus present, the minimum detectable rank-biserial correlation at 80% power is 0.212; the observed value is 0.097. So a moderate or larger association is excluded, while a small one is not. The report should not claim more than that.

One honest complication. Median FASN rises monotonically across invasion grades (262.7 → 308.7 → 326.1 TPM), which is the direction the manuscript predicted. The means do not (423 → 434 → 383), the test is null, and the effect is below the detectable threshold. We mention the trend because omitting it would be selective, not because it supports the claim.

3 Audit

Check	Verdict	Finding
Positive control	**Strong**	$\rho = 0.667$ between two independent AFP assays; join and matrix verified
Power for the null	**Quantified**	Observed effect 0.097 vs minimum detectable 0.212 — small effects not excluded
Censored inputs	**Handled**	43 of 238 serum AFP values are '>' or '<' capped; rank-based tests are unaffected in ordering
Expression sanity	**Plausible**	FASN 12.9–2763.5 TPM, median 304.6, no zeros
Gates	**6 / 6 PASS**	Sample count, gene presence, clinical join, control, replication

4 Consequence for the HCC manuscript

The revised manuscript should report the lipogenic co-expression phenotype and state plainly that FASN showed no association with stage, vascular invasion, grade, survival status or recurrence in either of two independent cohorts, with the power limitation stated. Combined with the earlier finding that FASN protein is *lower* in tumour while JUN protein is higher, and that no resolvable locus exists for the transcript designated HELC, the honest position is a descriptive biomarker report with an explicitly negative mechanistic result.

5 How to verify

`evidence/fasn_by_sample.csv` carries FASN TPM, AFP TPM and serum AFP for all 238 samples, so the positive control and every FASN test can be recomputed. Also included: `config.yaml` (hash `f4da7e3d...`, written before any outcome was read), `gates.json`, `results.json` (all p and q values), `audit_power.json`, `inputs.json` (file hashes), `MANIFEST.sha256`, `env.txt`, `m3_hcc_fasn.py`.

Run directory `runs/20260806T225833Z-R03hcc-3f8bece`. Data: Chinese Liver Cancer Atlas via cBioPortal (`hcc_clca_2024`), Open Database Licence. No patient-level identifiers reproduced.